

EMBER: A 100 MHz, 0.86 mm², Multiple-Bits-per-Cell RRAM Macro in 40 nm CMOS with Compact Peripherals and 1.0 pJ/bit Read Circuitry

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Abstract—Compact and energy-efficient resistive RAM (RRAM) macros are challenging to design due to large read/write circuits that decrease storage density, low-conductance cells that increase read latency, and routing parasitics that increase read energy costs. Multiple-bits-per-cell RRAM can boost bit density, but to our knowledge, no prior demonstrations fully integrate the read/write circuitry needed for multiple-bits-per-cell storage. This work presents a multiple-bits-per-cell RRAM macro, which (1) demonstrates read/write circuit compaction through constrained optimization of driver and pass gate transistor sizes, (2) introduces a common-mode bleed conductance at the sense amplifier inputs, reducing read settling time by 11.35× for low conductance cells, and (3) cuts read path capacitance to further reduce read access time and energy. It is the first embedded RRAM storage macro to achieve fully-integrated multiple-bits-per-cell readout and write-verification without any off-chip reference generation or sensing. The 0.86 mm² macro operates at 100 MHz with 64k×48 = 3M cells in 40 nm CMOS, achieving 1 b/cell read operation with 1.0 pJ/bit energy, and 2 b/cell read with 1.1 pJ/bit. Normalizing for process scaling, the macro demonstrates the highest effective RRAM cell density to date of 5.6e-3 b/F² for 1 b/cell and 1.3e-2 b/F² for 2 b/cell, an improvement of 31% and 204%, respectively, over best prior work.

Index Terms—Resistive RAM, Multiple-Bits-per-Cell, Multi-Level Cell (MLC), Embedded Memory, Non-Volatile Memory

I. INTRODUCTION

Resistive random access memory (RRAM) is a suitable on-chip memory technology for edge computing because of its back-end-of-line compatibility with CMOS, high unit cell density, multiple-bits-per-cell storage, non-volatility, and low-energy operation versus embedded flash [1]. Multiple-bits-per-cell RRAM ideally increases storage density by a factor of n (where $n = \log_2(\# \text{ of levels per cell})$) over single-bit-per-cell [2] but faces several challenges, illustrated in Fig. 1: (1) large read/write peripheral circuits attenuate storage density improvements, and the expanded conductance range for multiple-bits-per-cell operation can (2) slow down read and (3) increase average read energy. Lastly, to our knowledge, (4) no prior RRAM storage macro¹ integrates the circuitry needed for both multiple-bits-per-cell readout and write-verify operation.

In this paper, we present the **Efficient Multiple-Bits-per-cell Embedded RRAM** macro (EMBER, Fig. 2) that addresses the above-mentioned limitations by: (1) decreasing total bit line/source line (BL/SL) driver and pass gate area by 6.07× compared to [4] by using thinner oxide high-voltage transistors and trading off BL/SL write pass gate area with

¹Fully-integrated multiple-bits-per-cell operation with an RRAM array is demonstrated in [3], but the read/write circuits in that design are neither compact nor adjacent to the RRAM unit cells as is typical in a storage macro.

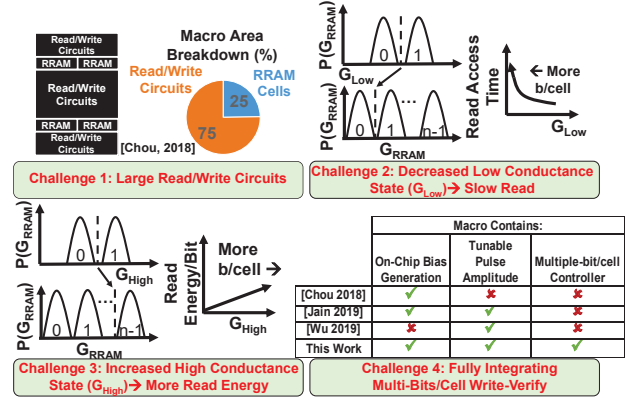


Fig. 1: Multiple-bits-per-cell RRAM macro design challenges.

BL/SL driver area, (2) reducing sense amplifier settling time for low-conductance RRAM cells by 11.35× with a common-mode bleed conductance, and (3) reducing read energy by 37% via intelligent placement of read access switches. EMBER demonstrates reliable 1-2 bits per cell on-chip write-verify and readout with an average read energy of 1.0 pJ/bit (on par with the state-of-the-art) for 1 b/cell and 1.1 pJ/bit for 2 b/cell. EMBER's normalized array density is 5.6e-3 b/F² for 1 b/cell and 1.3e-2 b/F² for 2 b/cell, representing an improvement of 31% and 204%, respectively, over the state-of-the-art [4].

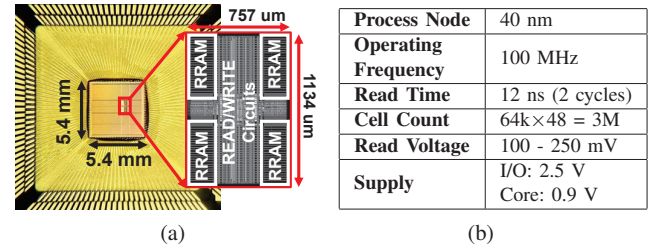


Fig. 2: Summary of EMBER macro. (a) Die photo of the EMBER chip and (b) overview of the macro specifications.

II. ARRAY & READ/WRITE CIRCUIT ARCHITECTURE

The EMBER macro architecture (Fig. 3) consists of four 0.75 Mcell RRAM unit cell array quadrants, divided by shared peripherals used for read/write. Each quadrant has 24 sub-arrays, each with 512 word lines (WLs) and 64 bit

lines (BLs) ($64 \times 512 \times 24 = 0.75\text{M}$ cells). Each unit cell contains one select transistor attached to a programmable resistor (1T1R). The select transistor gate connects to a WL, and its source connects to a source line (SL). The RRAM cell electrodes connect to a BL and select transistor drain. The RRAM arrays in EMBER employ a common SL architecture for improved cell density [4]. Each half of the array contains 24 sense amplifiers and write drivers, so a word of up to 48 cells can be read or programmed in parallel.

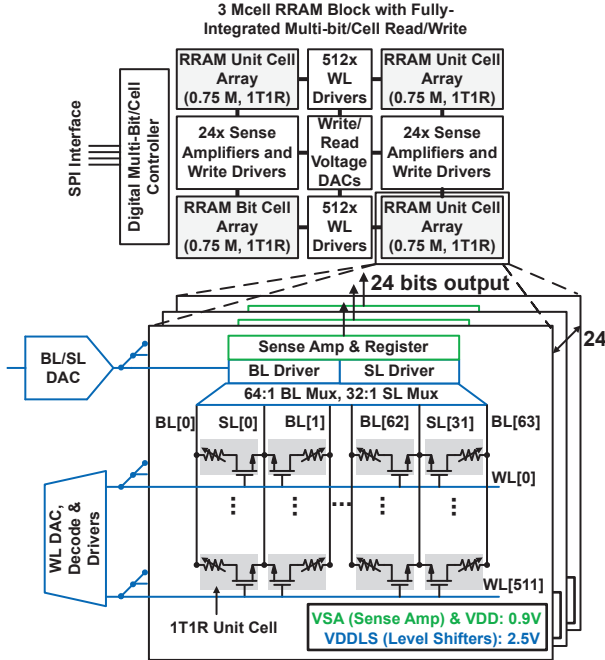


Fig. 3: EMBER top-level architecture and array design.

The array peripherals are divided into WL and BL/SL components. The input address is used to select a WL and drive a DAC-generated voltage on that WL. Each WL spans two quadrants of the array, and fine WL DAC voltage control imposes a soft current limit across RRAM cells during write. On the BL/SL, DACs adjust the applied write voltage and sense amplifiers enable read operations on the BL. Sense amplifiers compare a selected cell's conductance against a 6-bit user-controlled reference at a BL voltage specified by the read DAC. The read/write analog interfaces in EMBER are operated by a digital controller, which receives commands over a serial peripheral interface (SPI).

A. Technique 1: Read/Write Circuit Compaction through Constrained Optimization of Driver and Pass Gate Sizes

The total peripheral area (dominated by BL/SL pass gates and write drivers) is comparable to the area of the RRAM cells (e.g., Fig. 4a); this significantly lowers the effective bit density. The write path transistors must be wide enough to minimize IR drop when resetting high-conductance RRAM cells to a low conductance, but making peripheral transistors too large lowers macro density and slows read because of extra RC delay. We thus aim to minimize total peripheral area under the constraint that requisite SET/RESET voltages are met. We allow write transistor size variation with two relative sizing parameters (Fig. 4b): (1) the ratio between

the total SL-side and BL-side pass gate and driver transistor on-resistance (γ in Fig. 4b), and (2) the ratio between the total driver and the total pass gate on-resistance (λ in Fig. 4b).

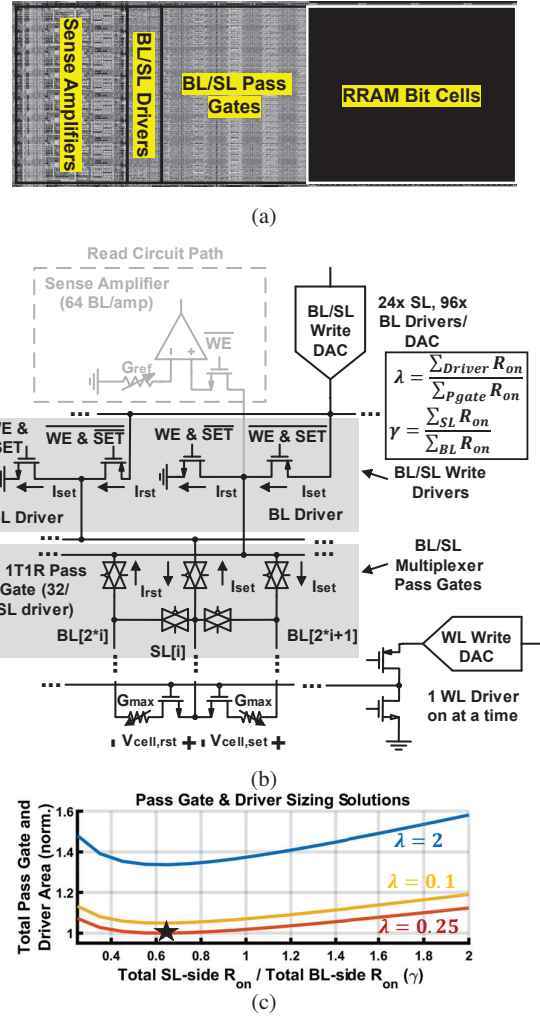


Fig. 4: (a) Read/write interface layout. (b) RRAM write path schematic. BL/SL write DAC current goes through the path starting at (1) the BL/SL driver (SET/RESET), (2) a pass gate to (3) induce voltage drop across a unit cell, and (4) an SL/BL (SET/RESET) pass gate and (5) driver to ground. (c) Parameter search to find the smallest write path design. Minimum area is achieved when $\lambda = 0.25$, $\gamma = 0.65$.

In each iteration of our (γ, λ) sweep, we obtain the minimum driver/pass gate transistor widths and total area by applying the unit cell SET/RESET voltage constraints. As shown in Fig. 4c, the global minimum area appears at $\lambda = 0.25$, $\gamma = 0.65$. The minimum is shallow with $\sim 10\%$ fluctuation over a wide γ and λ range. The use of this minimization technique coupled with choosing thinner oxide write path transistors resulted in a $6.07\times$ reduction in write circuit area compared to [4].

B. Technique 2: Faster Read Times with Bleed Conductance

EMBER uses an offset-cancelling current-mode sense amplifier design inspired by [5]. Our design is shown in

Fig. 5, with modifications to the prior design highlighted in red and blue. The red contributions are common-mode bleed conductances to limit sense amplifier settling time. The blue contribution is a read biasing circuit shared between 24 sense amplifiers in one half of the macro. The amplifier input settling time constant is described by:

$$\tau_{\text{settling}} = \frac{C_{BL}nV_t}{V_{\text{read}}(G_{RRAM} + G_{\text{bleed}})} \quad (1)$$

Above, C_{BL} is the input-referred capacitance on the BL side of the amplifier, $n = 1.5$ (ratio of MOSFET sub-threshold slope to 60 mV/decade), $V_t = \frac{k_B T}{q}$, V_{read} is the cell read voltage, G_{RRAM} is the cell conductance, and G_{bleed} is a common-mode bleed device conductance (Fig. 5a, red). When using this design with low G_{RRAM} and no G_{bleed} , the settling time for accurate comparisons increases due to a $1/G_{RRAM}$ dependence. G_{bleed} forces 99% read current settling to occur by $T_{\text{max}} \approx 4\tau_{\text{settling}}$, and the sense amplifier clock frequency becomes $\frac{1}{2T_{\text{max}}}$. G_{bleed} detaches the choice of RRAM conductance operating range from the read frequency specification, allowing for low G_{RRAM} in a multiple-bits-per-cell scheme.

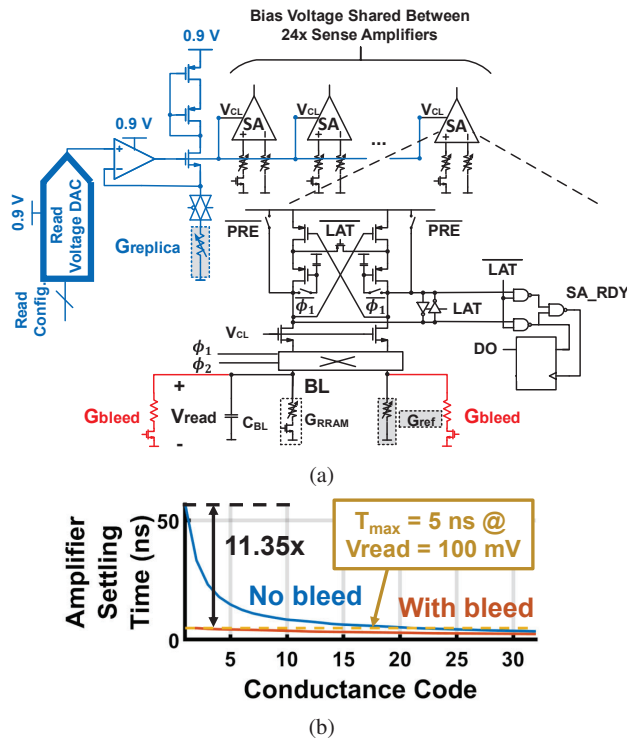


Fig. 5: (a) Offset-cancelling current-mode sense amplifier design (inspired by [5]), with modifications highlighted in red and blue. Operation waveform is similar to that in [5]. A shared circuit generates V_{CL} . (b) Simulation of settling times. Minimum G_{bleed} that meets $T_{\text{max}} = 5$ ns (100 MHz) is used.

To achieve additional energy savings, we share the bias voltage (V_{CL}) generation circuit (Fig. 5a, blue) with the 24 sense amplifiers present in each half of the BL/SL interface and turn off sense amplifiers at the earliest time that G_{RRAM} is determined. The shared reference reduces sense

amplifier area by 60% and read energy by 30% compared to a design that incorporates local feedback amplifiers [6]. The conductance readout uses a ramping reference scheme in which the 6-bit G_{ref} linearly increases between read steps, resulting in up to $2^n - 1$ steps for n -bit storage. A sense amplifier that determines G_{RRAM} early saves energy by idling after detecting a change in amplifier data output (DO).

C. Technique 3: Reducing Sense Amplifier Load Capacitance via Smart Placement of Read Access Switches

Since the high-voltage device pitch in the BL/SL driver/pass gate is larger than the RRAM unit cell pitch, the pass gates are physically stacked at the array edge (Fig. 6). The read/write access routes to the pass gates create a branching wire structure originating at a shared sense amplifier. Rather than a single large read access switch, each branch has a dedicated smaller read access switch, limiting the wire and pass gate capacitance at the sense amplifier input. This change reduces (1) settling time by 20% compared to attaching a sense amplifier with a single branch $T_{\text{max}} = 5$ ns to all branches and (2) read energy by 37% compared to increasing G_{bleed} to meet $T_{\text{max}} = 5$ ns with all branches loading the sense amplifier input.

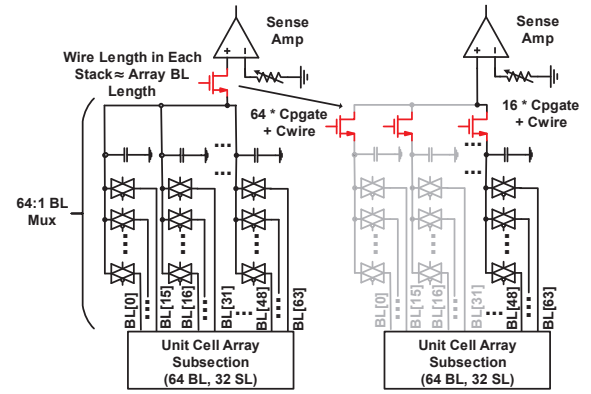


Fig. 6: Schematic of pass gate stacking from Fig. 4a. Duplicating the read access switch allows cell reading without driving excess capacitance, decreasing read energy and settling time.

D. Multiple-Bits-per-Cell Write-Verify

We design a configurable on-chip write-verify controller containing a register file with settings that define the READ/WRITE conductance thresholds and pulse amplitude settings for each level (up to 16 levels, 4-bits-per-cell). It enables low-level (single-bit) commands, such as SET, RESET, SENSE, along with high-level (multiple-bits-per-cell) memory commands READ, WRITE, and CYCLE. During WRITE (demonstrated in Fig. 7), we primarily use pulse amplitude modulation on the WL/BL/SL to reduce write access time [2], [7], though write pulse width is also configurable. We demonstrate 1 b/cell through 4 b/cell storage, with conductance ranges allocated via sigma-based allocation (described in [2]). While 1-2 b/cell storage shows low bit error rate (BER), further work is needed to bring 3-4 b/cell BER down to usable levels for on-chip storage.

III. RESULTS & DISCUSSION

A. Read/Write Energy & Read/Write Area Breakdown

The average read energy using the full reference conductance range (4 - 256 μS) varies from 1.0 pJ/bit for 1 bit/cell

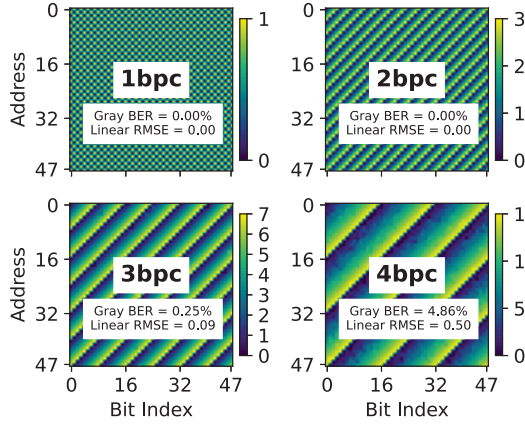


Fig. 7: Checkerboard patterns programmed to demonstrate multiple-bits-per-cell capability. Cell readout is performed 10 seconds after programming. 48 words (with 48 bits each) are shown for 1-4 bits per cell. Gray BER: bit error rate using Gray coding (relevant for digital data storage). Linear RMSE is root mean square error of a programmed symbol under linear coding (relevant for analog data storage, e.g., [8]).

operation to 2.2 pJ/bit for 4 bit/cell operation (Fig. 8). The sense amplifier supply (VSA) consumes 50% of the total per-bit read energy at this design's 100 MHz operating frequency. The average read energy per bit scales non-linearly with unit cell storage resolution since on average $\frac{2^{n-1}}{n}$ sense amplifier reads are performed per bit of information in an n -bit scheme. The macro quiescent power draw is 5 μ W, excluding the digital controller and using off-chip power gating for the write DAC supply. During SET/RESET, power is drawn mainly from the write DAC supply, varying from ~ 2 mW to ~ 50 mW for 48 cells. This corresponds to between ~ 0.4 pJ/cell and ~ 10 pJ/cell per SET/RESET pulse for a 10 ns pulse width (the minimum at 100 MHz). The power draw is highly dependent on the conductance of the cells being programmed as well as the settings of the write DACs. The endurance of the macro was measured to be $>10^6$ for 1 bit per cell. Finally, the macro area breakdown is presented in Fig. 9a, indicating that the BL/SL write drivers and multiplexers dominate the peripheral area.

B. Comparison with Other Macro Designs

A comparison between EMBER and the prior state-of-the-art is given in Table I. EMBER expands the Pareto front,

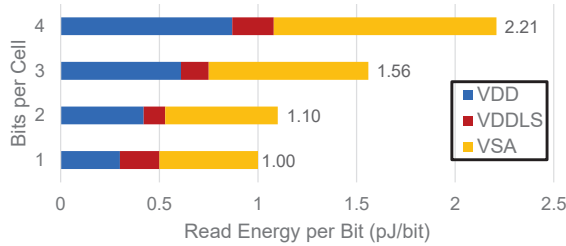


Fig. 8: Measured average read energy breakdown at 100 MHz operation with $V_{read} = 100$ mV. Read was repeatedly performed on checkerboard patterns (see Fig. 7) with word address scrolling. Average read power was measured via source-measure unit. VDD: core digital supply (0.9 V), VDDLs: level shifter supply (2.5 V), VSA: sense amplifier supply (0.9 V).

offering the highest effective density (normalized for process technology), improving by 31% over the state-of-the-art for 1 b/cell and 204% for 2 b/cell. At the same time, the macro matches the state-of-the-art in read energy per bit (Fig. 9b). 3-4 b/cell implementations show promise for further density increase, but BER needs to be brought down for practical use in digital applications.

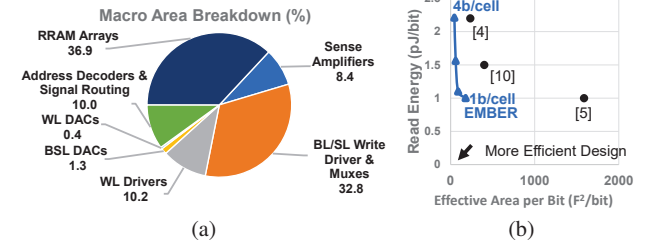


Fig. 9: (a) Macro area breakdown by percentage. (b) Read energy vs. unit cell area for EMBER and prior RRAM macros.

	This Work	[9]	[5]	[10]	[4]
Integrated Read/Write	Yes (1-4 b/cell)	No	Yes (1 b/cell)	Yes (1 b/cell)	Yes (1 b/cell)
Process Node	40 nm	130 nm	22 nm	22 nm	40 nm
Capacity (Mcells)	3	0.016	3.6	13.5	11
Multi-Bits/Cell	1-4 b/cell	1-2.3 b/cell	None	None	None
Read Energy (pJ/bit)	1.0 (1 b/cell) 1.1 (2 b/cell)	1.8 (1 b/cell)	1.0* (1 b/cell)	1.5 (1 b/cell)	2.2 (1 b/cell)
Read Time (1 b)	12 ns	23 ns	10 ns	10 ns	9 ns
Density (bit/F ²)	5.6e-3 (1 b/cell) 1.3e-2 (2 b/cell)	6.8e-5† (1 b/cell) 1.4e-4† (2.3 b/cell)	6.3e-4†	2.5e-3†	4.27e-3†

*Pre-silicon simulation, † Layout-estimated

TABLE I: Comparison with state-of-the-art RRAM macros.

IV. CONCLUSION

In this work, we introduce the EMBER RRAM macro which contains: (1) BL/SL access circuitry scaled to maximize density while providing adequate write strength, (2) bleed conductance to bound read circuit settling time and (3) read access switch placement to reduce read path capacitance. Finally, (4) EMBER demonstrates up to 2 b/cell write-verification with low BER using on-chip read/write bias circuitry and controllers, and 3-4 b/cell shows early promise. The read energy matches prior state of the art at 1.0 pJ/bit for 1 b/cell (1.1 pJ/bit for 2 b/cell), and the process-normalized storage density improves by 31% (204% for 2 b/cell). EMBER demonstrates techniques for efficient RRAM array design and is the first fully-integrated multiple-bits-per-cell RRAM storage macro.

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